Jian Feng

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Education

University of Waterloo Sep 2023 - Aug 2027
Bachelor's degree, Computer Science, Finance, and Statistics *Waterloo, ON, Canada*

Skills

Languages:	Python, C/C++, Java, C#, Go, Bash, SQL
Backend & Systems:	.NET, Spring Boot, FastAPI, Node.js
Databases:	PostgreSQL, MySQL, MongoDB, Redis
Cloud & DevOps:	AWS, GCP, Azure, Docker, Kubernetes, Terraform, GitHub Actions, Linux, Jenkins

Experience

Manulife Financial Corporation Toronto, ON, Canada
Software Engineer *May 2026 – Aug 2026*

- Owned automation of **1K+ daily cross-system operations** on an internal financial-services platform serving thousands of developers; ServiceNow pre-validation and skip-on-failure handling removed the manual exception queue.
- Strengthened access control and auditability by replacing plaintext basic-auth GitHub access with **GitHub App token exchange** (PyGithub); added team-scoped validation and least-privilege permissions across catalog automations.
- Added AKS cluster and certificate checks to GOCC, enabling a **Foundry LLM agent to compare GitOps PRs with live cluster state** and flag deployment misconfigurations before they reached production.

Treasury Board Secretariat Toronto, ON, Canada
Software Developer *Sep 2025 – Dec 2025*

- Independently built the job-title normalization module for a **5K-firm public dataset** (.NET MVC + SQL Server), cutting **85%** of the administrative backlog that had blocked peer-group and governance reporting.
- Rebuilt stored procedures, added indexes, and removed cursors, cutting report page load **30%** for analyst-facing reviews of 5K firms.
- Led the migration of **30 Selenium tests across five suites** to Katalon on Azure DevOps CI/CD; adoption by other QA teams simplified their no-code onboarding.

Lansi Institute of Artificial Intelligence in Medicine Remote
Data Analyst *May 2024 – Dec 2024*

- Owned a parallelized genomic ETL pipeline for **~100GB of multi-omics data**, using per-sample `multiprocessing` fan-out to cut preprocessing **89%, from days to hours**.
- Applied XGBoost and SMR/MR² across a **68-study T2D meta-analysis**, narrowing the candidate drug-target shortlist used in published papers.

Sanyang Technology Co., Ltd. Guangdong, China
Software Engineer *Jun 2024 – Nov 2024*

- Built a **.NET financial-reconciliation module** that automated variance tracking and removed manual month-end close checks.
- Integrated REST APIs with Kafka and Redis to ingest high-throughput sales data, powering **real-time revenue dashboards** for business users.

Projects

MiniKV | Go

- Implemented a Go LSM key-value store with a CRC32C write-ahead log, skip-list MemTable, compressed SSTables, sparse indexes, and **Bloom filters that skip disk reads for negative lookups**.
- Added snapshot reads and commit-time read-set validation for **serializable optimistic concurrency**, plus asynchronous leader/follower replication over HTTP NDJSON and gRPC with sequence-based recovery after reconnects.

wlp4-compiler | C++

- Built a **multi-pass C++ compiler** that scans, LR-parses, and type-checks WLP4 before generating ARM64 assembly; code generation supports control flow, procedures, pointers, and arrays.
- Reduced generated assembly size by **79.6%** versus reference `wlp4c` across 65 programs; CI checks byte-identical output on a 25-program regression suite.

Achievements

- **Velocity Co-op Problem Award** (top submission in cohort); Moonshot Hackathon Winner.
- Grew UX Research Hub **32× (10→318 members)** as Director; secured a Google UXR keynote at no cost.

Coursework

Compiler Design, Algorithms, Parallel/Concurrent/Distributed Programming, Database Design & Operational BI